

CHAPTER 3 WRITING GUIDE

What is Chapter 3- DESIGN AND METHODOLOGY

Chapter 3 explains **how you built your system and how you will evaluate it**. Think of it as your blueprint and your plan. Everything you write here should answer one question:

"How exactly did you design, build, and plan to test your system?"

Write everything in **future tense** since this is a proposal. Use **"will be"**, **"shall"**, **"are to be"** — not **"was"** or **"is."**

3.1 Requirement Analysis and Documentation

What it is: A list of everything your system must do and the data it needs.

What to write:

Requirements Gathering Methods Table Explain how you found out what the system needs. Common methods:

- Interview — talk to the staff or manager of your partner org
- Observation — watch how they currently do things manually
- Document Analysis — review their existing records, forms, receipts
- Dataset Profiling — check if their historical data is usable for your algorithm

Put these in a table: Method | Purpose | Participants/Sources

Functional Requirements Table List what the system must do. Format:

- FR-01, FR-02, FR-03...
- Each one has a Name and a Description
- Start each description with "The system shall..."
- Include at least one FR specifically for your algorithm (*e.g., FR-06: ARIMA Sales Prediction — The system shall generate forecasts...*)

Non-Functional Requirements Table List how well the system must perform.

Format:

- NFR-01, NFR-02...
- Categories: Performance, Usability, Security, Reliability, Accuracy, Compatibility, Scalability

Data Requirements Table ← Describe the dataset your algorithm needs:

- Where it comes from
- What fields are required
- Minimum number of records
- Data quality standards
- How access was authorized (Data Sharing Agreement)

3.2 Conceptual Framework

What it is: A diagram that shows the big picture of your system using the Input-Process-Output (IPO) model.

What to write:

- Insert your IPO diagram as Figure 1
- Write a narrative that describes:
 - **INPUT** — what data and user actions go into the system
 - **PROCESS** — two layers:
 - System Layer (inventory, records, transactions)
 - Analytics Layer (data cleaning, algorithm, prediction) ← DBA exclusive
 - **OUTPUT** — what the system produces (dashboard, reports, forecasts, alerts)

Common mistake: DBA students forget to describe both layers. Your IPO must show the analytics pipeline, not just the web system.

3.3 Development Methodology

What it is: The framework or process you followed to build the system.

What to write:

- Choose one SDLC: **Agile**, Waterfall, or RAD — and explain *why* you chose it
- Add **CRISP-DM** for the analytics component
- Make a phases table with columns:
 - Phase Name
 - Agile SDLC Activities
 - CRISP-DM Activities

- Key Deliverable

Why CRISP-DM? It is the standard process for data mining and machine learning projects. It covers: Business Understanding → Data Understanding → Data Preparation → Modeling → Evaluation → Deployment. This maps directly to how your algorithm was developed.

3.4 System Design

What it is: All the diagrams that show how your system is structured and how it works.

Rule: Every diagram must be followed by a narrative paragraph explaining what it shows.

Diagram	What it Shows	Description
3.4.1 Context Diagram	Existing system vs. proposed system. Who interacts with the system and what data flows in and out.	Show how the analytics module receives data
3.4.2 Data Flow Diagram Level 0	The entire system as one process with major data flows	Standard
3.4.3 Use Case Diagram	All actors and what they can do in the system	Include analytics-specific use cases: Trigger Prediction, View Forecast, View Model Performance
3.4.4 ERD	Database tables, fields, and relationships	Must include analytics tables: ForecastRecord, ModelPerformanceLog
3.4.5 System Architecture	Three-tier structure: Frontend → Backend → Database	Must show how the algorithm is called and where it lives (e.g., Flask API)
3.4.6 Flowchart	Step-by-step process flows	Must include: Data Preprocessing flowchart AND Algorithm Execution flowchart
3.4.7 Mockups	High-fidelity screen designs with actual colors and layout	Must include Analytics Dashboard and Data Upload screens

For mockups, write a description under each figure explaining:

- What is on the screen
- What the user can do on it
- How analytics outputs are displayed
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Diagram	Top Recommendation	Alternative
ERD	draw.io	DBDiagram.io

Use Case Diagram	draw.io	StarUML
Context Diagram	draw.io	Lucidchart
DFD Level 0 / Level 1	draw.io	Visual Paradigm
System Architecture	draw.io	Canva
Flowchart	draw.io	Lucidchart
High-Fidelity Mockups	Figma	Adobe XD

3.5 Tools, Technologies, and Development Environment

What it is: A table of every tool or technology you used and why you chose it.

Format: Category | Tool/Technology | Version | Justification

Required categories

Category	Examples
Frontend	React.js, Vue.js, Bootstrap
Backend	Laravel, Django, Node.js
Database	MySQL, PostgreSQL
Python / ML Libraries	Python, statsmodels, scikit-learn, Pandas, NumPy
Model Serving	Flask API, FastAPI
Jupyter Notebook	JupyterLab
Data Visualization	Chart.js, Plotly.js
Version Control	Git, GitHub
IDE / Code Editor	VS Code, PyCharm
Diagram Tools	draw.io, Lucidchart
UI/UX Design	Figma
Deployment	XAMPP, cPanel

Tip: If you used a free or open-source tool, still include it and justify why you chose it over alternatives.

3.6 Development Cost

What it is: An estimated breakdown of how much the system cost to build.

Why it matters: It shows the economic value of your system to the partner organization.

Required cost categories:

Category	How to Compute
Human Resource	Total hours per member × ₱75/hr (or current rate)
Software & Licensing	List all tools; ₱0.00 if open-source

Hardware	Laptop depreciation for project duration
Dataset Acquisition	Cost of dataset if purchased
Cloud Computing	Cost of GPU/cloud if used
Internet & Utilities	₱500/month × members × months
Documentation & Printing	Manuscript, binding, instruments
Transportation	Trips to partner org × fare
Contingency	10% of subtotal
TOTAL	Sum all above

Important: End this section with one paragraph summarizing the total cost and explaining why it is low (e.g., open-source tools, no paid dataset, local training).

3.7 Evaluation Methods

What it is: Your plan for testing and measuring the quality and acceptability of your system.

3.7.1 Evaluation Using ISO/IEC 25010

What it is: A technical quality evaluation conducted by IT professionals.

What to write:

- **Purpose** — why you are using this standard
- **Quality Characteristics** — the 7 characteristics you will evaluate: Functional Suitability, Performance Efficiency, Usability, Reliability, Security, Maintainability, Portability
- For each characteristic, describe what it means *for your specific system*
- **Evaluators** — minimum 3 IT professionals, purposive sampling
- **Instrument** — 5-point Likert scale evaluation form
- **Process Overview** — how evaluators will access and assess the system

3.7.2 Evaluation Using TAM

What it is: A user acceptance evaluation conducted by actual end-users.

What to write:

- **Purpose** — why TAM was chosen
- **TAM Constructs:**

- PU (Perceived Usefulness) — does the system help them do their job better?
- PEOU (Perceived Ease of Use) — is it easy to use?
- BI (Behavioral Intention) — will they use it regularly?
- **For DBA:** TAM items must reference the analytics features (*e.g.*, *"The sales forecast helps me decide when to reorder stock"*)
- **Respondents** — 15 to 30 end-users, purposive sampling
- **Instrument** — 5-point Likert scale questionnaire

3.7.3 Data Handling and Analysis

This is the most structured subsection. Follow this exact order:

A. Respondents and Population Description

State the total population (N) and break it down by role. Identify the two evaluator groups:

1. End-user respondents for TAM — 15 to 30, purposive sampling
2. IT professional evaluators for ISO/IEC 25010 — minimum 3, purposive sampling

B. Data Analysis

Write these three things in order:

1. Describe the 5-point Likert scale:

5 – Strongly Agree, 4 – Agree, 3 – Neutral, 2 – Disagree, 1 – Strongly Disagree

2. Insert the **Likert Scale Conversion Table**:

Rating Range	Interpretation
4.51 – 5.00	Highly Acceptable
3.51 – 4.50	Acceptable
2.51 – 3.50	Moderately Acceptable
1.51 – 2.50	Slightly Acceptable

1.00 – 1.50	Not Acceptable
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- Write a narrative paragraph describing what each range means for your specific system.

C. Statistical Treatment of Data

State that the **Arithmetic Mean (AM)** is used. Present the formula:

$$AM = \Sigma(w_i f_i) / n$$

Define each variable:

- AM = Arithmetic Mean
- w_i = Likert scale weight (5, 4, 3, 2, 1)
- f_i = Frequency of responses for each weight
- n = Total number of respondents

Explain that AM is computed per item, per quality characteristic, per TAM construct, and for the overall score.

D. Statistical Tools Used

Present three tools in this order:

Frequency — used to count how many respondents chose each rating per item.

Percentage — used to express the proportion of responses. Formula:

$$P = (F \times 100) / N$$

Where: P = Percentage, F = Frequency, N = Total respondents

Weighted Mean — connects back to the AM formula. This is the main tool for interpreting results.

E. Algorithm Performance Metrics

The performance of the embedded algorithm will be evaluated independently from the system quality assessment conducted under ISO/IEC 25010 and the user acceptance evaluation conducted under TAM. While the ISO/IEC 25010 and TAM evaluations measure the quality and acceptability of the web-based system as a whole, the algorithm performance evaluation specifically measures how accurately and reliably the analytical model produces its outputs. The metrics used depend on the type of algorithm integrated into the system.

For Regression-Based Algorithms *Used when the algorithm predicts a continuous numerical value*

Examples: ARIMA, Linear Regression, Multiple Regression, SVR, Prophet

Applicable systems: sales forecasting, demand prediction, price prediction, enrollment forecasting, resource consumption prediction

For Classification-Based Algorithms *Used when the algorithm predicts a category, label, or class*

Examples: Random Forest, Decision Tree, SVM, KNN, Naïve Bayes, Logistic Regression, Neural Networks for classification

Applicable systems: disease diagnosis prediction, student performance classification, customer churn detection, spam detection, fraud detection, crop disease classification, sentiment analysis

Metric	Formula
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$
Precision	$TP / (TP + FP)$
Recall (Sensitivity)	$TP / (TP + FN)$
F1-Score	$2 \times (Precision \times Recall) / (Precision + Recall)$

Variable Definitions:

- **TP** (True Positive) — predicted Positive; actually Positive
- **TN** (True Negative) — predicted Negative; actually Negative
- **FP** (False Positive) — predicted Positive; actually Negative (*false alarm / Type I Error*)
- **FN** (False Negative) — predicted Negative; actually Positive (*missed case / Type II Error*)

Confusion Matrix

The Confusion Matrix is presented alongside the classification metrics as a complete summary of the model's prediction results. It breaks down all predictions into four categories — TP, TN, FP, FN — allowing the research team to identify not just overall accuracy but specifically where the model is making errors and what type of errors are most frequent.

The confusion matrix will be presented as a labeled table or heatmap visualization in Chapter IV alongside the classification metrics table.

For Clustering-Based Algorithms *Used when the algorithm groups unlabeled data into natural clusters without a predefined target class*

Examples: K-Means, DBSCAN, Hierarchical Clustering, Gaussian Mixture Models

Applicable systems: customer segmentation, student profiling, product categorization, geographic zone grouping

Which Metrics Apply to Your Study

Algorithm Type	Required Metrics
Regression / Forecasting (ARIMA, Linear Regression, SVR, Prophet)	MAE, MAPE, RMSE
Classification (Random Forest, Decision Tree, SVM, KNN, Naïve Bayes, Logistic Regression)	Accuracy, Precision, Recall, F1-Score + Confusion Matrix
Clustering (K-Means, DBSCAN, Hierarchical)	Silhouette Score, Davies-Bouldin Index
Neural Networks / Deep Learning	Use regression or classification metrics depending on the task

Important Note for Students

Present only the metrics that correspond to your specific algorithm type. If your system uses a **regression algorithm**, report MAE, MAPE, and RMSE only. If it uses a **classification algorithm**, report Accuracy, Precision, Recall, F1-Score, and the Confusion Matrix. If it uses a **clustering algorithm**, report the Silhouette Score and Davies-Bouldin Index. Never present all metric types in a single study — only include what is directly applicable to the algorithm your system uses.

End with a statement that the complete algorithm performance results — including confusion matrix visualization for classification systems, or actual-versus-predicted charts for regression systems — will be presented and interpreted in **Chapter IV**.